MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL TTL, SHIFT REGISTERS,
MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-38510/9D, dated 4 June 1980, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, second sentence: Delete "Three" and substitute "Two".

PAGE 2

1.3, Thermal resistance, delete and substitute:

"Thermal resistance, junction to case, (θ_{JC}):
Cases A, B, C, D, E and F 1/ - - - - (See appendix C of MIL-M-38510)"

1.3: After "Junction temperature" add "2/".

1.4: Delete "Ambient" and substitute "Case"

Bottom of page, add the following footnote:

"1/ When a thermal resistance value is included in appendix C of MIL-M-38510, it shall supersede the value stated herein.

2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883."

PAGE 3

1.4, device type 06: Delete "27" and substitute "32"; delete "20" and substitute "25".

* 2.1, delete and substitute the following:

"Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence."

* 3.4: Delete "ambient" and substitute "case".

PAGES 4, 5, 6, and 7

TABLE I, under conditions: Add "$-55^{\circ}C \leq T_C \leq +125^{\circ}C$". TABLE I, V_{IC} conditions: Delete "T_A" and substitute "T_C".

PAGE 5

TABLE I, T_{PLH1}, T_{PHL1}, and T_{PLH2}, T_{PHL3} tests for device type 02, minimum limits column: Delete "12" and substitute "8" (4 places).

PAGE 7

TABLE II, Group A test requirements, class B devices; add "8".

TABLE II: Delete class C device column in its entirety.

PAGE 17

FIGURE 3: Delete and substitute new figure 3 as printed on page 5 of this amendment.

PAGE 21

NOTE 2, delete and substitute:

"2. R VALUES ARE EITHER 4K, 5K, OR 6K."

PAGE 28

FIGURE 4: Delete and substitute new figure 4 as printed on page 6 of this amendment.

PAGE 38

FIGURE 4: Delete and substitute new figure 4 as printed on page 7 of this amendment.

PAGE 48

* FIGURE 6, Notes 1, 2, 3 and 4: Delete "T_A" (6 places) and substitute "T_C".

PAGE 54

* Notes 1 and 2: Delete "T_A" and substitute "T_C".

PAGE 57

* FIGURE 9: Delete and substitute new figure 9 as printed on page 8 of this amendment.

PAGE 58

FIGURE 9, Clock to output (t_{PHL2} and t_{PLH2}), note 1: Delete "t_{SETUP2} = 30 ns" and substitute "t_{SETUP2} = 34 ns".

PAGE 61

* Notes 2 and 3: Delete "T_A" and substitute "T_C".

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PAGE 62

* Note 2: Delete "T_A" and substitute "T_C".

PAGES 63 AND 64

* FIGURE 11: Delete and substitute new figure 11 as printed on pages 9 and 10 of this amendment.

PAGE 67

* TABLE III, device 01, delete and substitute new page as printed on page 11 of this amendment.

PAGES 65 THROUGH 90

* TABLE III, heading: Delete "(pins not designated are open, high level or low level)" and substitute ("pins not designated may be high $\geq$ 2.0 V or low $\leq$ 0.8 V, or open)".

* TABLE III: Delete "T_A" and substitute: "T_C" for the entire table.

PAGES 70 AND 71

TABLE III, device type 02, tests 125 through 144 and 146 through 165, minimum limits column: Delete "12" and substitute "8".

PAGE 76

Footnote 7/: Add "For schematics incorporating a 4.5 kΩ base resistor, the minimum and maximum limits shall be -0.6 and -1.5 mA, respectively."

Footnote 8/, first sentence: Delete "C". Add new sentence, "For schematic circuit C, the minimum and maximum limits shall be -0.6 and -1.5 mA, respectively."

PAGE 79

TABLE III, test 89, pin 2: Delete "B" and substitute "A".

PAGE 84

TABLE III, device type 05, delete and substitute new table III as printed on page 12 of this amendment.

PAGE 87

TABLE III, device type 06, test 10, column 11: Delete "-.8 mA" and substitute "16 mA".

PAGE 91

* 4.2a: Add "or E".

4.2c, delete in its entirety and substitute the following:

"c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510."

* 4.4, delete and substitute the following:

"4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspection (see 4.4.1 through 4.4.4)."

* 4.4.2, add after MIL-STD-883: "Electrical parameters shall be as specified in TABLE II herein."

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PAGE 92

4.4.3b, delete and substitute:

"b. Subgroups 3 and 4 shall be added to group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A."

* 4.4.3c: Add "or E".

* 4.5, delete and substitute the following: "4.5 Methods of inspection. Methods of inspection shall be specified as follows:"

4.5.2: Delete in its entirety.

4.6: Delete in its entirety.

6.1: Delete in its entirely.

6.3d: Delete "procuring" and substitute "the contracting".

6.3e: Delete in its entirety.

6.3h: Delete "Requirements for carriers, special" and substitute "Requirements for special carriers,".

PAGE 93

* 6.3: Delete "electrical ground (common terminal)" and substitute "ground zero voltage potential".

* 6.3, V_{IC}: Delete in its entirety.

6.6, Generic-industry type: Delete "7495", "7496", "74164", "74165", "74194", and "74195"

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Air Force - 17
Army - ER
Navy - EC

Review activities:
Air Force - 11, 19, 85, 99
Army - AR, MI
Navy - OS, SH, TD
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-0781)

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Device type 01

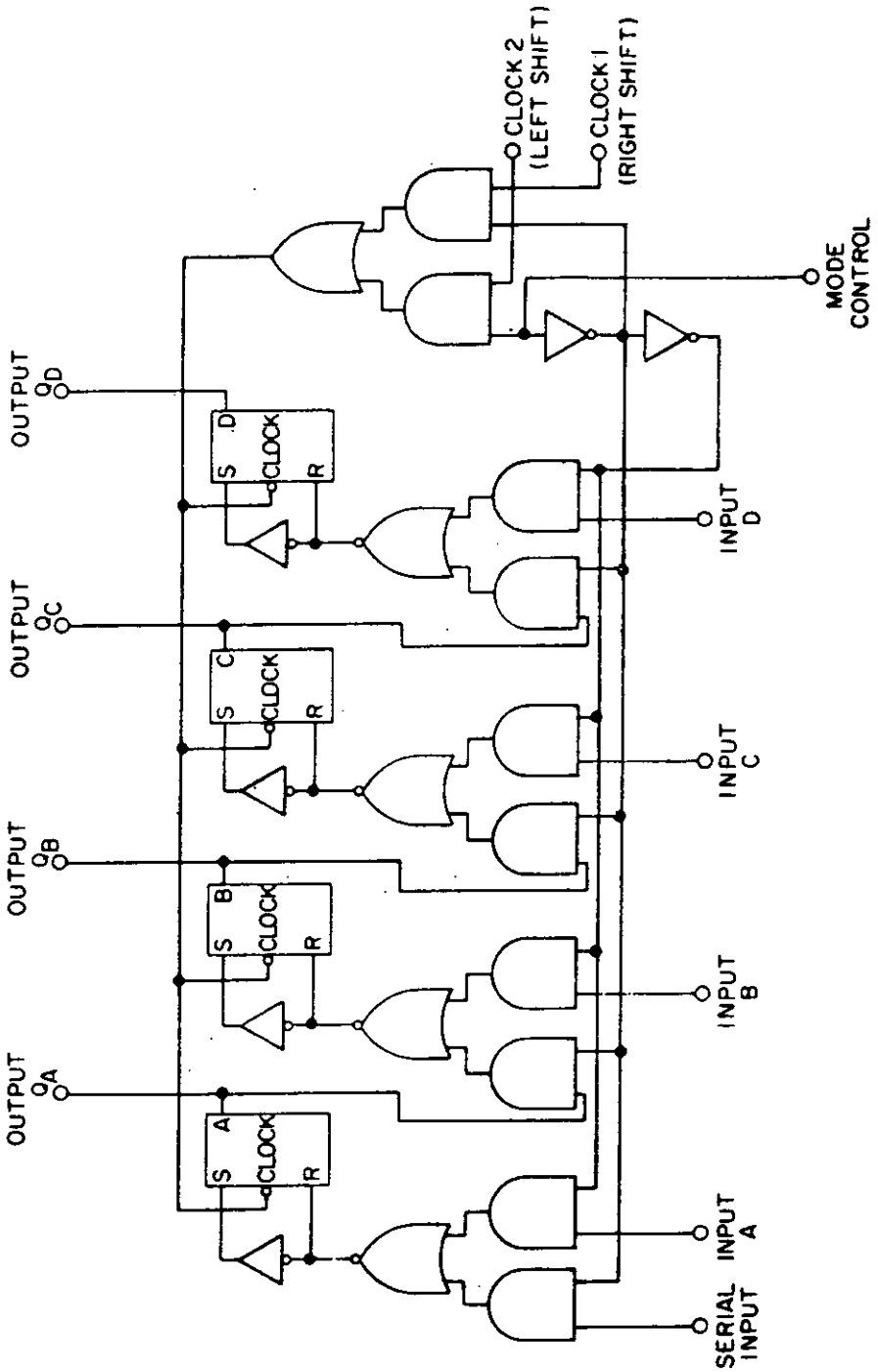


FIGURE 3. Logic diagrams.

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Device type 03
Circuit C

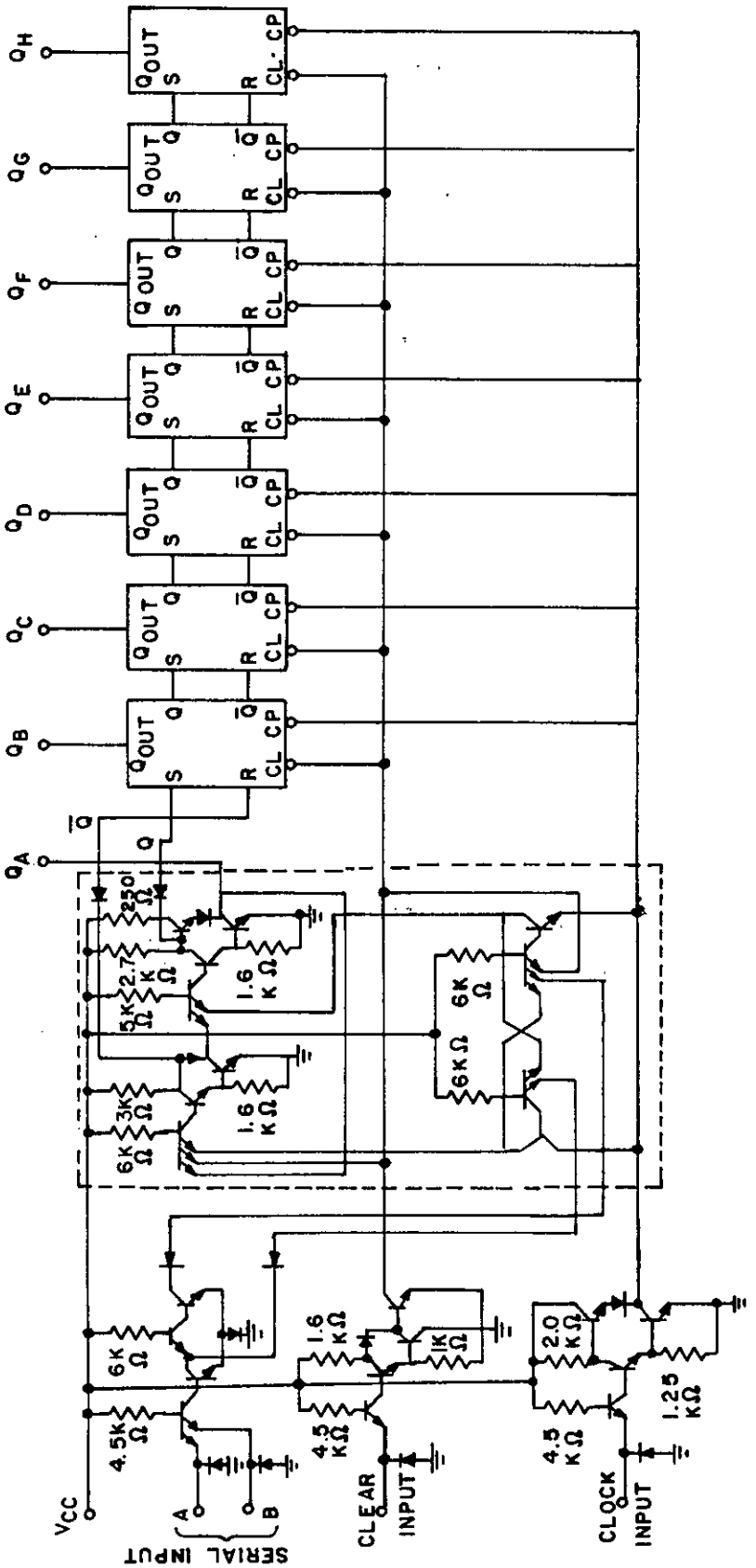
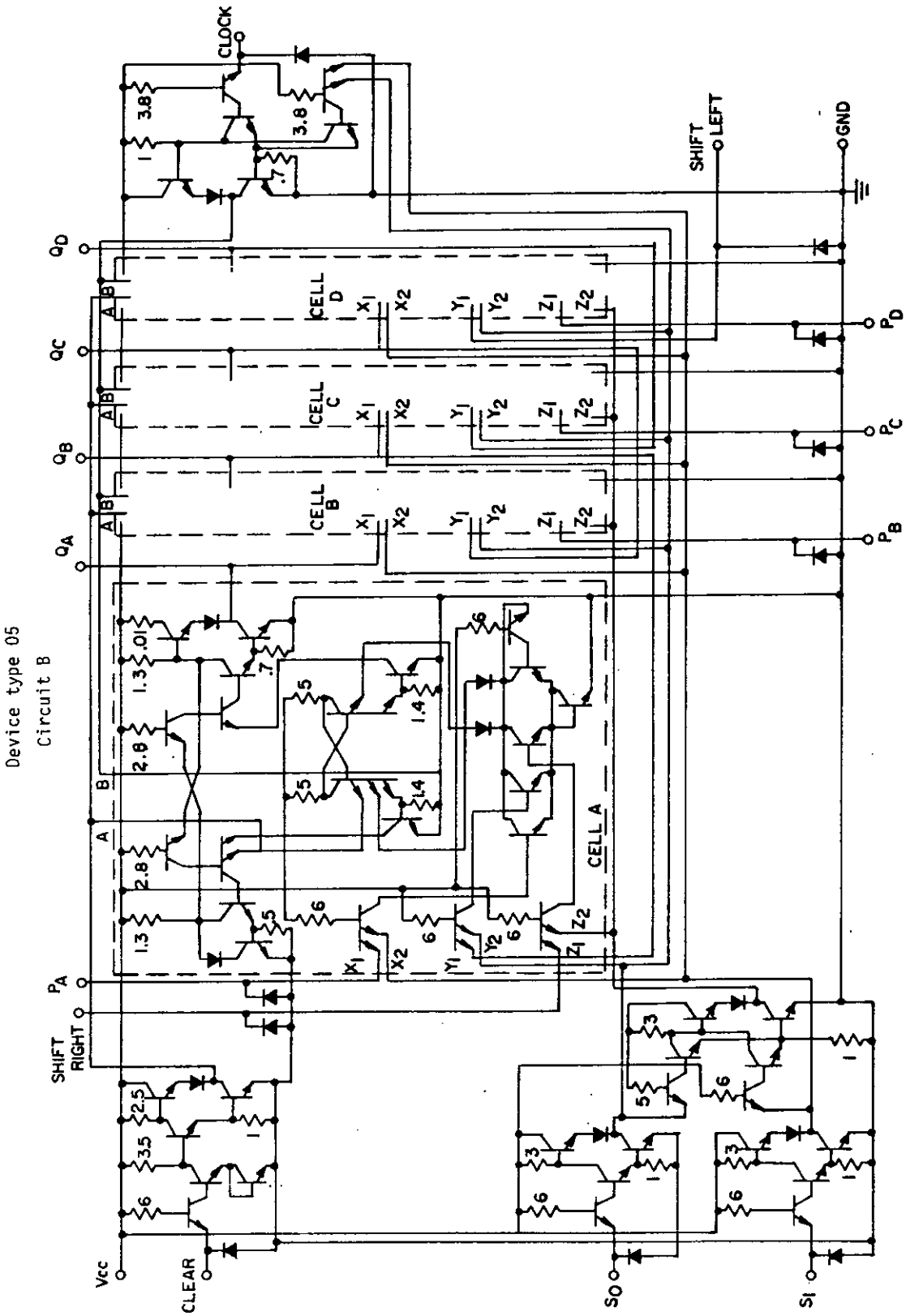


FIGURE 4. Schematic circuits - Continued.

Component values shown are normal.

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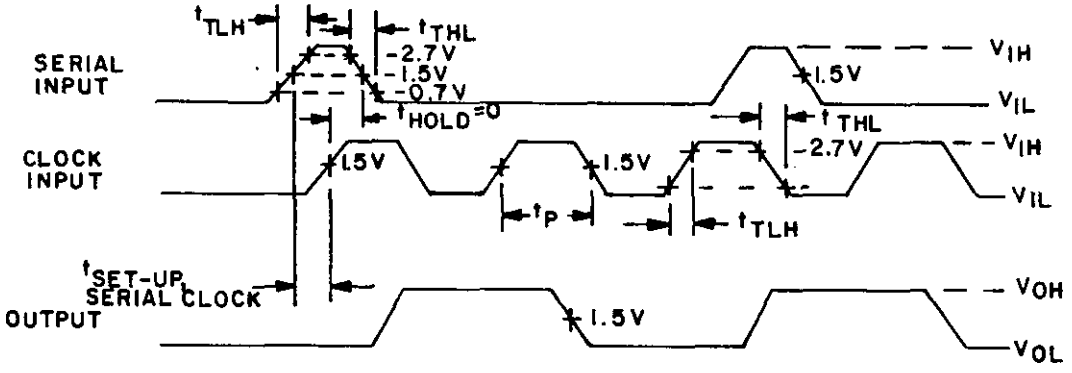


NOTES:
1. Component values shown are nominal.
2. All resistor values are in kΩ.

FIGURE 4. Schematic circuits - Continued.

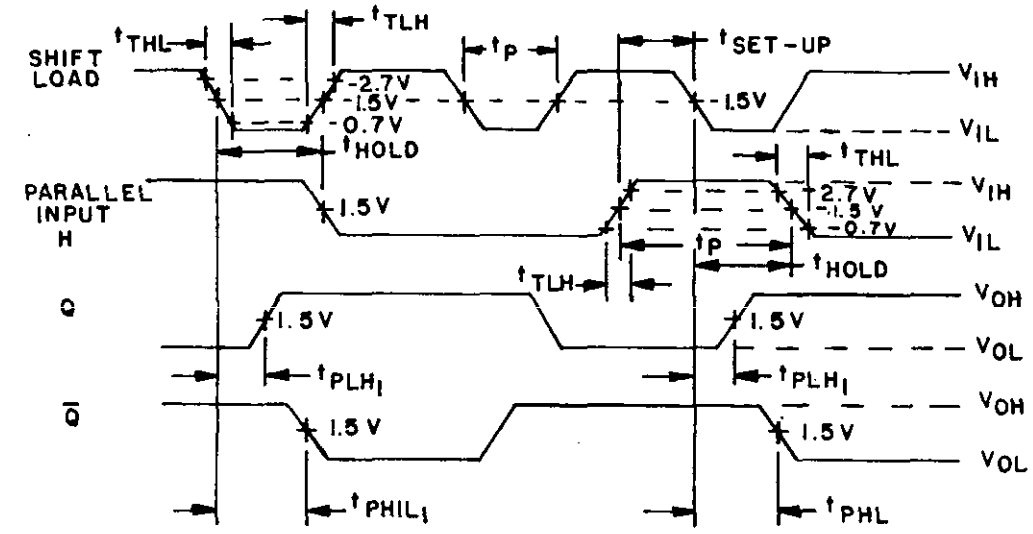
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MAXIMUM CLOCK FREQUENCY, f_{MAX}



NOTES:
1. Clock pulse characteristics: PRR = 18 MHz at $T_C = 25^oC$, PRR = 14 MHz at $-55^oC \leq T_C \leq 125^oC$, $t_{TLH} = t_{THL} \leq 10$ ns, $t_p = 20$ ns minimum.
2. Serial pulse characteristics: PRR = 9 MHz at $T_C = 25^oC$, 7 MHz at $-55^o \leq T_C \leq 125^oC$ $t_p = t_{SETUP} + t_{HOLD}$, $t_{SETUP} = 35$ ns minimum, $t_{HOLD} = 0$, $t_{TLH} = t_{THL} \leq 5$ ns.
3. Shift load characteristics: $t_{TLH} = \leq 10$ ns, $t_{SETUP} = 45$ ns.
4. Clock inhibit = GND, A thru H = GND.

SHIFT LOAD TO OUTPUT (t_{PHL1}, t_{PLH1})

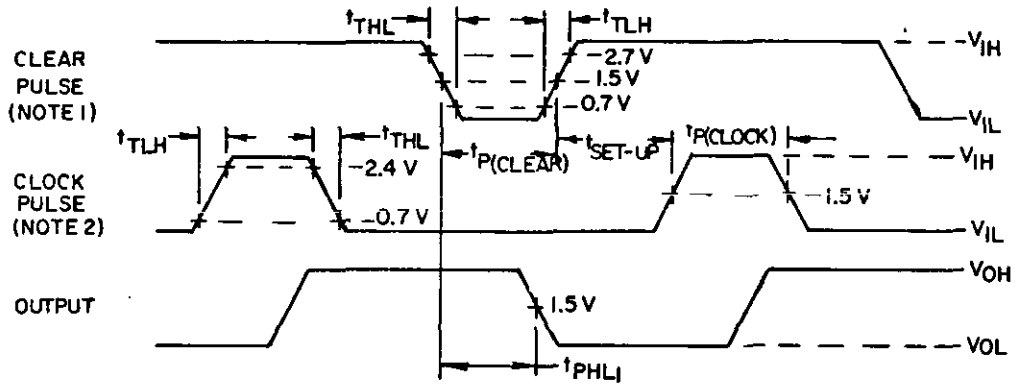


NOTES:
1. Shift load characteristics: PRR = 1 MHz, $t_p = 25$ ns, $t_{THL} = t_{TLH} \leq 10$ ns.
2. Parallel input characteristics: PRR = 500 kHz, $t_p = t_{SETUP} + t_{HOLD} = 40$ ns, $t_{SETUP} = 10$ ns, $t_{HOLD} = 30$ ns, $t_{THL} \leq 10$ ns.
3. Clock = clock inhibit = GND, A thru G = GND, serial = OPEN.

FIGURE 9. Switching test circuit and waveforms for device type 04 - Continued.

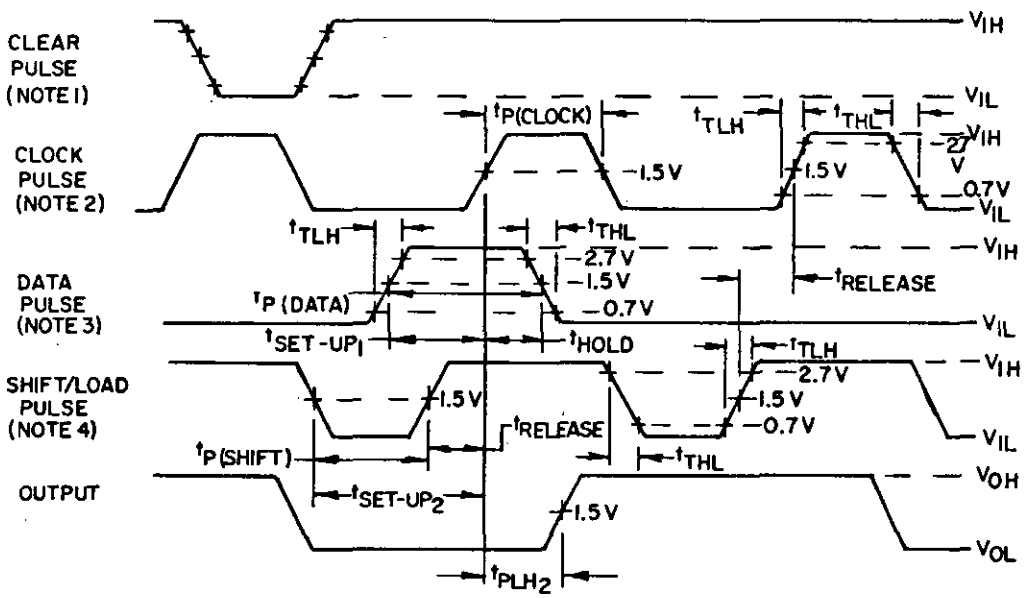
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CLEAR TO OUTPUT (t_{PHL1})



NOTES:
1. Clear pulse characteristics: $t_{P(CLEAR)}$ = 12 ns, t_{SETUP} = 25 ns, PRR = 1 MHz.
2. Clock pulse characteristics: $t_{P(CLOCK)}$ = 16 ns, PRR = 1 MHz.

CLOCK TO OUTPUT (t_{PLH2})

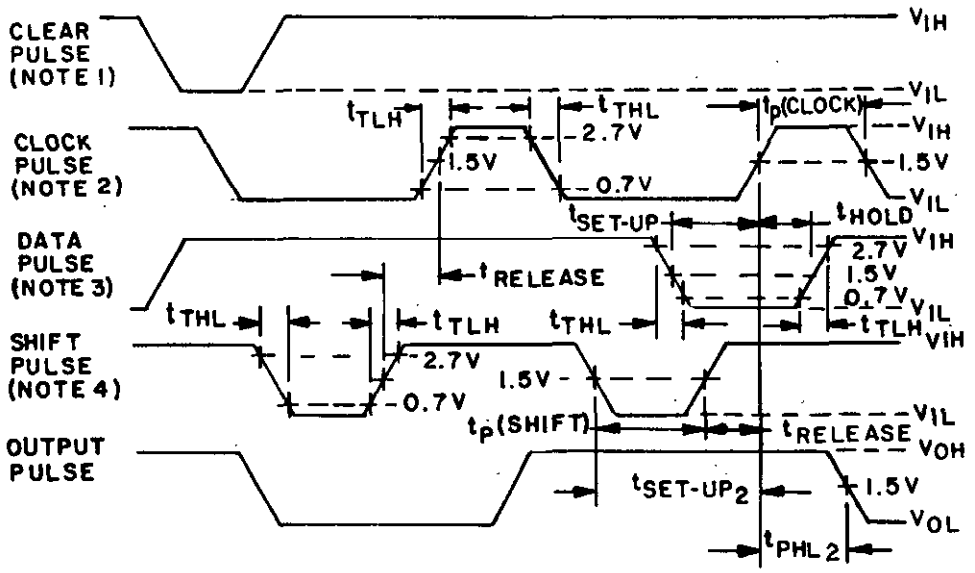


NOTES:
1. The clear pulse is a momentary GND, then V_{IH} is applied to the clear input.
2. Clock pulse characteristics: $t_{P(CLOCK)}$ = 16 ns, PRR = 2 MHz.
3. Data pulse characteristics: $t_{P(DATA)}$ = 25 ns, t_{SETUP1} = 25 ns, t_{HOLD} = 0, PRR = 1 MHz.
4. Shift/load pulse characteristics: $t_{P(SHIFT)}$ = 17 ns, $t_{RELEASE}$ = 10 ns, t_{SETUP2} = 27 ns PRR = 2 MHz.

FIGURE 11. Switching test circuit and waveforms for device type 06 - Continued.

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CLOCK TO OUTPUT (t_{PHL2})



NOTES:
1. The clear pulse is a momentary GND, then V_{IH} is applied to the clear input.
2. Clock pulse characteristics: $t_{P(CLOCK)}$ = 16 ns, PRR = 2 MHz.
3. Data pulse characteristics: $t_{P(DATA)} = t_{SETUP} + t_{HOLD}$ = 25 ns, t_{SETUP1} = 25 ns, t_{HOLD} = 0, PRR = 1 MHz.
4. Shift/load pulse characteristics: $t_{P(SHIFT)}$ = 22 ns, $t_{RELEASE}$ = 10 ns, t_{SETUP2} = 32 ns, PRR = 2 MHz.

FIGURE 11. Switching test circuit and waveforms for device type 06 - Continued.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Test no.	1 Input serial	2 Input A	3 Input B	4 Input C	5 Input D	6 Mode control	7 GND	8 Clock 2	9 Clock 1	10 Q_D	11 Q_C	12 Q_B	13 Q_A	14 V_{CC}	Measured terminal	Test limits Min	Max	Unit
7 T_C = 25°C 4/ 7/	Truth table test 5/	3014	75	B	A	B	A	B	A	GND	B	B	L	L	L	L	4.5 V				
		"	76	"	A	B	A	B	"	"	A	"	"	L	"	L	"				
		"	77	"	A	B	A	B	"	"	B	"	"	H	"	H	"				
		"	78	"	A	B	B	A	"	"	B	"	"	H	"	H	"				
		"	79	"	B	"	B	"	"	"	A	"	"	H	"	H	"				
		"	80	"	B	"	A	"	"	"	B	"	H	L	H	L	"				
		"	81	"	A	"	B	"	"	"	B	"	"	L	"	L	"				
		"	82	"	A	"	A	"	"	"	A	"	"	L	"	L	"				
		"	83	"	A	"	A	"	"	"	B	"	"	H	"	H	"				
		"	84	"	B	B	B	B	"	"	B	"	"	H	"	H	"				
		"	85	"	B	B	B	B	"	"	A	"	"	H	"	H	"				
		"	86	"	B	B	B	B	"	"	B	"	L	L	L	L	"				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																				
9 T_C = 25°C	f_{MAX}	(Fig 6)	87	GND					GND	GND	IN	IN	OUT				5.0 V	Q_D	11		MHz
	t_{PLH}	3003 (Fig 6)	88	"	IN	IN	IN	IN	5.0 V	"	"	GND	OUT	OUT	OUT	OUT	"	Q_A	10	30	ns
		"	89	"					"	"	"	"					"	Q_B	"	"	"
		"	90	"					"	"	"	"					"	Q_C	"	"	"
		"	91	"					"	"	"	"					"	Q_D	"	"	"
	t_{PHL}	"	92	IN					GND	"	GND	IN	OUT	OUT	OUT	OUT	"	Q_A	"	35	"
		"	93	"					"	"	"	"					"	Q_B	"	"	"
		"	94	"					"	"	"	"					"	Q_C	"	"	"
		"	95	"					"	"	"	"					"	Q_D	"	"	"
10 T_C = 125°C	f_{MAX}	(Fig 6)	96	GND					GND	"	IN	"	OUT				"	Q_D	8		MHz
	t_{PLH}	3003 (Fig 6)	97	"	IN	IN	IN	IN	5.0 V	"	"	GND	OUT	OUT	OUT	OUT	"	Q_A	10	42	ns
		"	98	"					"	"	"	"					"	Q_B	"	"	"
		"	99	"					"	"	"	"					"	Q_C	"	"	"
		"	100	"					"	"	"	"					"	Q_D	"	"	"
	t_{PHL}	"	101	IN					GND	"	GND	"	OUT	OUT	OUT	OUT	"	Q_A	"	49	"
		"	102	"					"	"	"	"					"	Q_B	"	"	"
		"	103	"					"	"	"	"					"	Q_C	"	"	"
		"	104	"					"	"	"	"					"	Q_D	"	"	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

1/ A = normal clock pulse, except for subgroups 7 and 8 (see 4/).
2/ For device type 01, with schematics incorporating a 4 kΩ base resistor, the minimum and maximum limits shall be -0.5 and -1.4 mA, respectively. For schematics incorporating a 5 kΩ base resistor, the minimum and maximum limits shall be -0.5 and -1.4 mA, respectively. For schematics incorporating a 6 kΩ base resistor, the minimum and maximum limits shall be -0.4 and -1.3 mA, respectively.
3/ For device type 01, with schematics incorporating a 4 kΩ base resistor in the mode control input circuit, the minimum and maximum limits shall be -1.4 and -3.2 mA, respectively. For schematics incorporating a 5 kΩ base resistor, the minimum and maximum limits shall be -1.0 and -2.8 mA, respectively. For schematics incorporating a 6 kΩ base resistor in the mode control input circuit, the minimum and maximum limits shall be -0.8 and -2.6 mA, respectively.
4/ For subgroups 7 and 8, A = V_{CC}, B = GND, and X = indeterminate.
5/ The tests in subgroups 7 and 8 shall be performed in the sequence specified.
6/ Output voltages shall be either:
(a) H = 2.4 V minimum and L = 0.4 V maximum when using a high speed checker double comparator or
(b) H >1.5 V and L<1.5 V when using a high speed checker single comparator.
7/ Only a summary of attribute data is required.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883-method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				Clear	Shift right	Input A	Input B	Input C	Input D	Shift left	GND	S_0	S_1	Clock	Q_D	Q_C	Q_B	Q_A	V_{CC}				
1 T_C = 25°C	I_{OS}	3011	49	5.5 V		5.5 V	5.5 V	5.5 V	5.5 V		GND	5.5 V	5.5 V	A				GND	5.5 V	Q_A	-20	-57	mA
		"	50	"		"	"	"	"		"	"	"	"					"	Q_B	"	"	"
		"	51	"		"	"	"	"		"	"	"	"	GND	GND	GND		"	Q_C	"	"	"
		"	52	"		"	"	"	"		"	"	"	"					"	Q_D	"	"	"
	I_{CC}	3005	53	"	5.5 V	GND	GND	GND	GND	5.5 V	"	"	"	"					"	V_{CC}		63	"
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = 25°C 3/ 6/	Truth table test 5/	3014	54	B 2/	B 2/	A 1/	B 2/	A 1/	B 2/	B 2/	GND	A 1/	A 1/	A 1/	L	L	L	L	5.0 V		4/		
		"	55	A	"	"	"	"	"	"	"	"	"	A	"	L	L	L	"				
		"	56	"	"	"	"	"	"	"	"	"	"	B	"	L	H	L	"				
		"	57	"	"	"	"	"	"	"	"	"	"	A	"	H	H	H	"				
		"	58	"	"	B	A	B	A	"	"	"	"	A	"	H	H	H	"				
		"	59	"	"	B	B	B	"	"	"	"	"	B	"	H	L	L	"				
		"	60	"	"	B	B	B	"	"	"	"	"	A	"	H	L	L	"				
		"	61	"	"	A	"	A	"	"	"	"	"	A	"	H	L	L	"				
		"	62	"	"	"	"	"	"	"	"	"	"	B	"	H	L	L	"				
		"	63	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	64	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	65	"	"	"	B	"	B	"	"	"	"	B	"	H	L	L	"				
		"	66	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	67	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	68	B	"	"	"	"	"	"	"	"	"	B	"	H	L	L	"				
		"	69	A	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
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		"	72	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	73	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	74	"	A	B	B	B	"	B	"	"	"	B	"	H	L	L	"				
		"	75	"	A	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	76	"	A	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
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		"	82	"	A	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	83	"	B	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	84	"	B	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	85	"	A	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	86	"	A	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	87	"	A	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	88	"	B	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	89	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	90	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	91	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	92	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	93	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	94	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	95	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	96	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	97	"	"	"	"	"	"	"	"	"	"	A	"	H	L	L	"				
		"	98	"	"	"	"	"	"	"	"	B	"	A	"	"	"	"	"				

See footnotes at end of device type 05.

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